

Fast Adapted Delay and Sum Reconstruction Algorithm in Circular Photoacoustic Tomography

Soheil Hakakzadeh

Department of Electrical Engineering
Sharif University of Technology
Tehran, Iran

<https://orcid.org/0000-0002-6791-1031>

Seyed Masood Mostafavi

Department of Electrical Engineering
Sharif University of Technology
Tehran, Iran

mostafavi.seyedmasoud@ee.sharif.edu

Zahra Kavehvash

Department of Electrical Engineering
Sharif University of Technology
Tehran, Iran

kavehvash@sharif.edu

Abstract— Circular photoacoustic tomography (C-PAT) arrangements are widely used in PAT systems. Many algorithms are proposed for PAT to reconstruct images such as delay and sum (DAS), universal back-projection (UBP), etc., all suffering from their own disadvantages. In this paper, to overcome these limitations, we proposed a fast uniform adapted delay and sum (ADAS) and filtered adapted delay and sum (F-ADAS) for detectors with finite and infinite bandwidth, respectively. Moreover, the edge uniformity index (EUI) metric was proposed for quantitative evaluation. Two cases 1) infinite bandwidth and 25 dB SNR level and 2) 2.25 MHz, 80% bandwidth with the same SNR level were considered for numerical study. The simulation results indicate that by using ADAS and F-ADAS, EUI improves 17.3-39% for infinite bandwidth scenario and EUI value can reach the value of 1 in finite bandwidth scenario. Therefore, highly uniform reconstructed images can be thus obtained with our proposed algorithm.

Keywords— Delay and sum algorithm, Universal back-projection, Circular scanning photoacoustic tomography, Edge uniformity index, Reconstruction algorithm.

I. INTRODUCTION

Photoacoustic imaging (PAI) is one of the nascent imaging methods in the field of biomedical imaging (clinical and pre-clinical imaging applications). The idea of which was completed and introduced in the first decade of the last century by Wang [1]. Photoacoustic is a combination of photo (light irradiation) and acoustic (acoustic detection), that can take advantage of high optical resolution due to the high tissue absorption coefficient and high acoustic contrast [2]. This process consists of four steps (see Fig. 1). 1) irradiating tissue by laser (Fig. 1(a)). 2) Tissue warming due to light energy (Fig. 1(b)). 3) Producing an acoustic wave due to the heating and detecting the acoustic wave (PA signal) by the transducer/detector (Fig. 1(c)). 4) Post-processing and image reconstruction algorithms (Fig. 1(d)).

Photoacoustic tomography (PAT) can be used in different configurations depending on the application such as linear, circular, spherical and limited-view circular geometries [3, 4]. Linear geometries can be used for palm and forearm imaging [5], hand-held systems and mouse brain imaging [6]. The circular and spherical arrangement can be used for breast imaging [7, 8], animal whole-body imaging [9, 10], in-vivo three dimensional PAI [11, 12] and limited-view geometry for circular PAT can be used for thyroid and carotid imaging [13].

Many algorithms are used for reconstruction, such as delay and sum (DAS) and universal back-projection (UBP) [14] algorithms, which are the most widely used in PAT, due to their simplicity and their fast performance. through using these algorithms, video-rate and real time imaging can be obtained. One of the main shortcomings of DAS method is the nonuniformity of the reconstructed image and also the occurrence of artifacts. Yet, its advantage in addition to its high-speed is the non-dependency of its performance on the noise level. The reconstructed image with UBP method, on the other side, is a uniform image, but very sensitive to noise at high noise levels, the signal-to-noise ratio (SNR) and contrast ratio (CR) of the image will be drastically reduced.

Many other methods have been presented to improve reconstructed image quality of ultrasound/photoacoustic images (resolution, SNR, CR). Delay-multiply and sum (DMAS) and filtered delay multiply and sum (FDMAS) were proposed to improve lateral resolution in linear array geometries [15]. Moreover, coherence weighting factors combined with circular arrangement PAT are proposed to enhance CR [16]. Some model-based reconstruction algorithms were proposed to reconstruct 3D images for PAT [17], the aim of which is reconstructing an accurate image with high structural similarity index measure (SSIM) and good uniformity. The disadvantage of all of these algorithms is the high reconstruction time and computational cost. Based on this limitation, with these algorithms real time and video-rate imaging cannot be obtained.

In this paper we propose a fast adapted delay and sum (ADAS) and filtered adapted delay and sum (F-ADAS) algorithm for circular PAT arrangement. By using ADAS/F-ADAS algorithms uniform and high contrast reconstructed image can be obtained. Our ADAS/F-ADAS algorithms are not sensitive to noise and high noise levels will not degrade the image quality. Moreover, ADAS/F-ADAS have advantages of both DAS and UBP algorithm and they can be an alternative to conventional methods (DAS/UBP).

In section 2 the theory of ADAS and F-ADAS are presented. Section 3 contains the simulation study and results. and Section 4 is the discussion and conclusion.

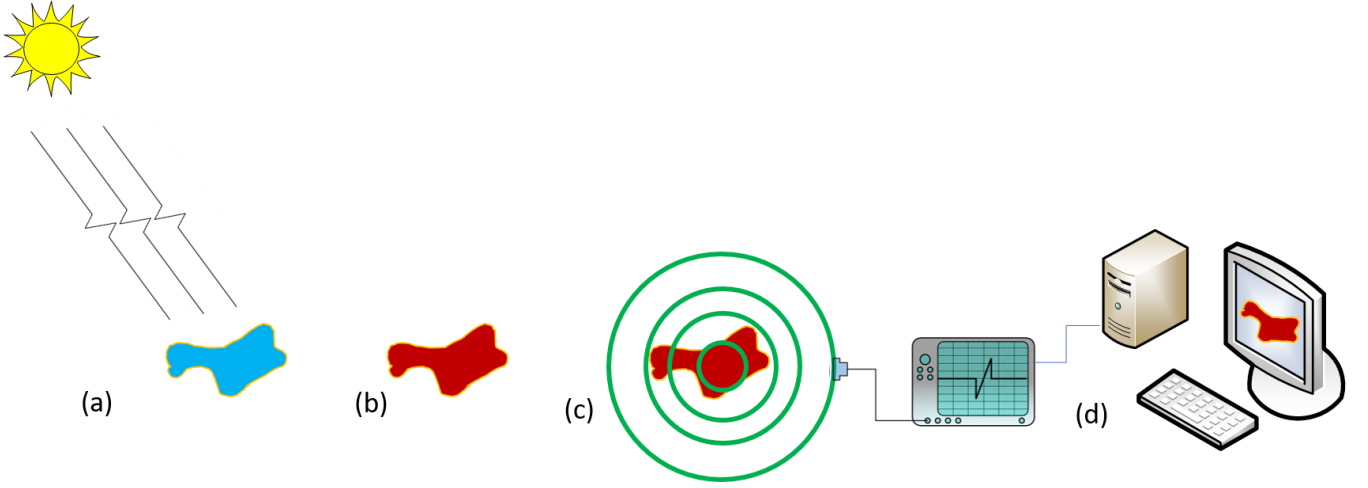


Fig. 1: schematic of photoacoustic phenomena. (a) light irradiates the PA absorber. (b) warming absorber due to the irradiation. (c) detecting the ultrasound wave (PA signal) generated by absorber. (d) post-processing and image reconstruction process.

II. MATERIALS AND METHODS

In photoacoustic phenomena, each absorber produces an initial pressure (PA wave). This pressure can be detected by a detector at distance x . The detected PA signal based on the initial pressure raised for a 2D geometry is given in (1).

$$p_x(t) = \frac{A \operatorname{Re}[e^{j(\omega t - kx)}] e^{-ax}}{\sqrt{x}} \quad (1)$$

Where, ω is the angular frequency, k is the wave-number, x is distance between absorber and detector which is tc , where c is the speed of sound (SOS) and t is the time of flight (TOF) of PA wave. The Re operator indicates the real part of the PA signal, A is the amplitude of initial pressure and a is attenuation coefficient. To compensate the attenuation caused by a coefficient and $\frac{1}{\sqrt{x}}$ in (1), time gain compensation (TGC) factor can be considered which is given in (2).

$$TGC(t) = \sqrt{ct} e^{act} \quad (2)$$

By multiplying (2) and (1), $p_{x,TGC}(t)$ can be obtained, which is given in (3).

$$p_{x,TGC}(t) = p_x(t) TGC(t) = A \operatorname{Re}[e^{j(\omega t - kx)}] \quad (3)$$

In DAS reconstruction procedure, each delayed PA signal is back-projected in the region of interest (ROI) and by summing all these PA signals, the final reconstructed image will be obtained.

In typical circular PAT arrangements, a transducer rotates around a chamber with scanning radius of R (see Fig. 2(a)). The indicated transducer, detects PA signals which are generated by absorber (Fig. 1(b, c)) within its field-of-view (FOV). Reconstruction in a circular geometry PAT can be interpreted as back projecting the captured ultrasonic rays by each transducer from each absorber point in the same angle. Based on that, an absorber can be divided into some angular

slices (Fig. 2(a)). Depending on the location of the absorber with respect to the scanning center, each angular slice of the absorber falls in the FOV of a number of transducers (assuming that the FOV of the transducers is large enough to accommodate the entire ROI). The number of transducers involved in the reconstruction of each angular slice is thus given as (4).

$$n_{\theta}^{(s)}(t) = r\theta/p \quad (4)$$

Where p , is the pitch of the array elements if ring array transducers are used or spatial step size if the used single element transducer. Moreover, r is the distance between absorber and scanning-center which is tc . It can be concluded from (4) that for 2 different angular slices (see 1, 2 in Fig. 2(b)) as we move away from the scanning center to the edge of the ROI, different angular slices of the absorber are reconstructed with different amplitudes, due to the different numbers of back-projected PA signals. To tackle this issue, each pixel in ROI is divided by the number of back-projected PA signals which is given in (5).

$$\begin{aligned} p_{x,ADAS}(t) &= p_x(t) \frac{TGC(t)}{n_{\theta}^{(s)}(t)} \\ &= A \operatorname{Re}[e^{j(\omega t - kx)}] p/r\theta \end{aligned} \quad (5)$$

From (5) it can be concluded that the intensity of each absorber is uniform in all angular slice.

One of the main disadvantages of the DAS algorithm is the presence of many DC components and artifacts in the final reconstructed image. To tackle this issue, a filtered adapted delay and sum (F-ADAS) is proposed here. In F-ADAS a DC block filter is applied on each back-projected A-line which is given in (6).

$$P_{x,F-DAS}(\omega) = P_{x,ADAS}(\omega) * \begin{cases} 1, & \omega \neq 0 \\ 0, & \omega = 0 \end{cases} \quad (6)$$

Where, $P_{x,F-DAS}(\omega)$ and $P_{x,ADAS}(\omega)$ are Fourier domain (FD) signals resulted from ADAS and F-ADAS algorithms, $*$ is multiplication operator and $\begin{cases} 1, & \omega \neq 0 \\ 0, & \omega = 0 \end{cases}$ is DC-block filter in FD. The summarized diagram of ADAS and F-ADAS algorithm are shown in Fig. 3.

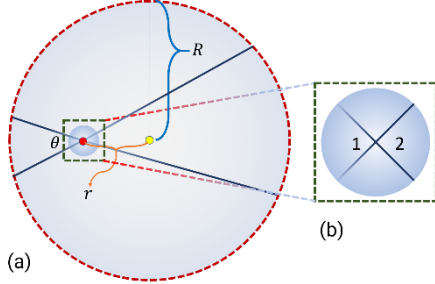


Fig. 2: (a) circular PAT arrangement with scanning radius R . red circle indicates a PA absorber, which its distance to the scanning radius is r . (b) zoomed area in (a). 2 different equal slices angles are indicated in (b)

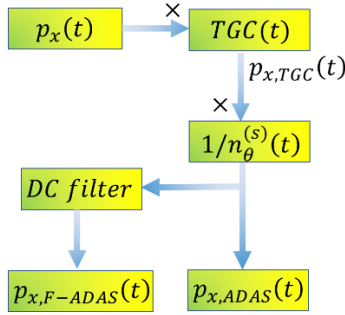


Fig. 3: Summarized diagram of ADAS/F-ADAS algorithms.

III. SIMULATION STUDY AND RESULTS

A. Simulation Study

k-wave toolbox [5] is used for simulation study. In these simulations, a 2D circular photoacoustic tomography geometry is considered. The medium is homogeneous and the speed of sound (SOS) is 1500 m/s. A full circular Derenzo phantom was used for all numerical results. 360-point transducers were placed on a circular chamber with 2.5 cm chamber radius. The medium grid size was 520×520 at both X-axis and Y-axis directions (5.2×5.2 cm). Perfect match layer (PML) with 20 grids were considered in both directions. All PA signals are detected by transducers with sampling frequency of 120 MHz and are stored in a desktop computer. Moreover, in all simulations, the attenuation coefficient is considered to be zero. The proposed method (ADAS and F-ADAS) is compared with DAS, TGC+DAS and UBP based on the reconstructed image quality.

In subsection B and C, Gaussian noise with SNR level 25 dB is added and the considered transducer in subsection B has infinite bandwidth and in subsection C, has central frequency of 2.25 MHz and 70% bandwidth. Moreover, for a better comparison of the reconstructed images, we set the negative parts of the reconstructed images in Fig. 4 to zero.

For quantitative comparison, edge uniformity index (EUI) which is given in (7) is reported for one-dimensional diagram of indicated dashed line (see yellow dashed-line in Fig. 4, 5 and 6).

$$EUI = 1 - \sqrt{\sum_{i=1}^2 (\mu_S - S_i)^2} \quad (7)$$

Where, μ_S is the mean value of the edge of the reconstructed image and S_i is the edge intensity, i . The closer the EUI value is to one, the more uniform the image will become. By substituting $(S_1 + S_2)/2$ in μ_S , we will reach to (8).

$$EUI = 1 - \sqrt{\frac{(S_2 - S_1)^2}{2}} \quad (8)$$

B. Case 1: Infinite bandwidth detector with 25 dB SNR level.

In Case 1, with 25 dB SNR level, UBP algorithm has more artifacts than other methods based on the derivative term in its algorithm (Fig. 4(b, e, h)). In DAS and DAS+TGC algorithms, for each point source, the intensity of the reconstructed image decreases together with the addition of a DC term, as the off-center distance increases (see dashed circle in Fig. 4(c, f, i, j, m, p)) which will cause a non-uniform reconstructed image for equal size of PA absorbers. The reason for these issues is the $n_\theta^{(s)}(t)$ effect, which is given in (4). In ADAS algorithm a uniform reconstructed image for equal size of PA absorbers is given. Still, due to the infinite bandwidth, here again a DC term is added to the image in reconstruction procedure which fortunately can be removed easily by a DC-filter before back-projection (F-DMAS algorithm). In F-DMAS, a uniform and high contrast reconstructed image can be obtained. Based on the 1D-line, shown in Fig. 5(a), it can be seen that reconstructed image with UBP is very sharp but artifacts and fluctuations are much more than other algorithms (see Fig. 5(b, c)). The reconstructed image with ADAS and F-ADAS is more uniform than others (see Fig. 5(b, c)). Quantitative values are indicated in Table 1.

It can be concluded from Table 1, that uniformity of reconstructed image based on ADAS and F-ADAS is about 0.95, which means that uniformity with these algorithms is approximately at its maximum value and is 17.3, 17.3 and 39 % better than UBP, DAS and DAS+TGC, respectively.

TABLE 1: Quantitative values (EUI) for infinite bandwidth and 25 dB SNR level scenario.

Algorithm	UBP	DAS	DAS+TGC	ADAS	F-ADAS
<i>EUI</i>	0.81	0.809	0.682	0.964	0.943

C. Case 2: 2.25 MHz central frequency, 80% bandwidth detector with 25 dB SNR level.

2.25 MHz central frequency transducer acts like a band-pass filter. Based on that, F-ADAS and ADAS results are the same (see Fig. 6). As indicated in Case 1 and based on the shown 1D-line, it can be seen that the reconstructed image with UBP has uniform edge with high variance artifacts. The

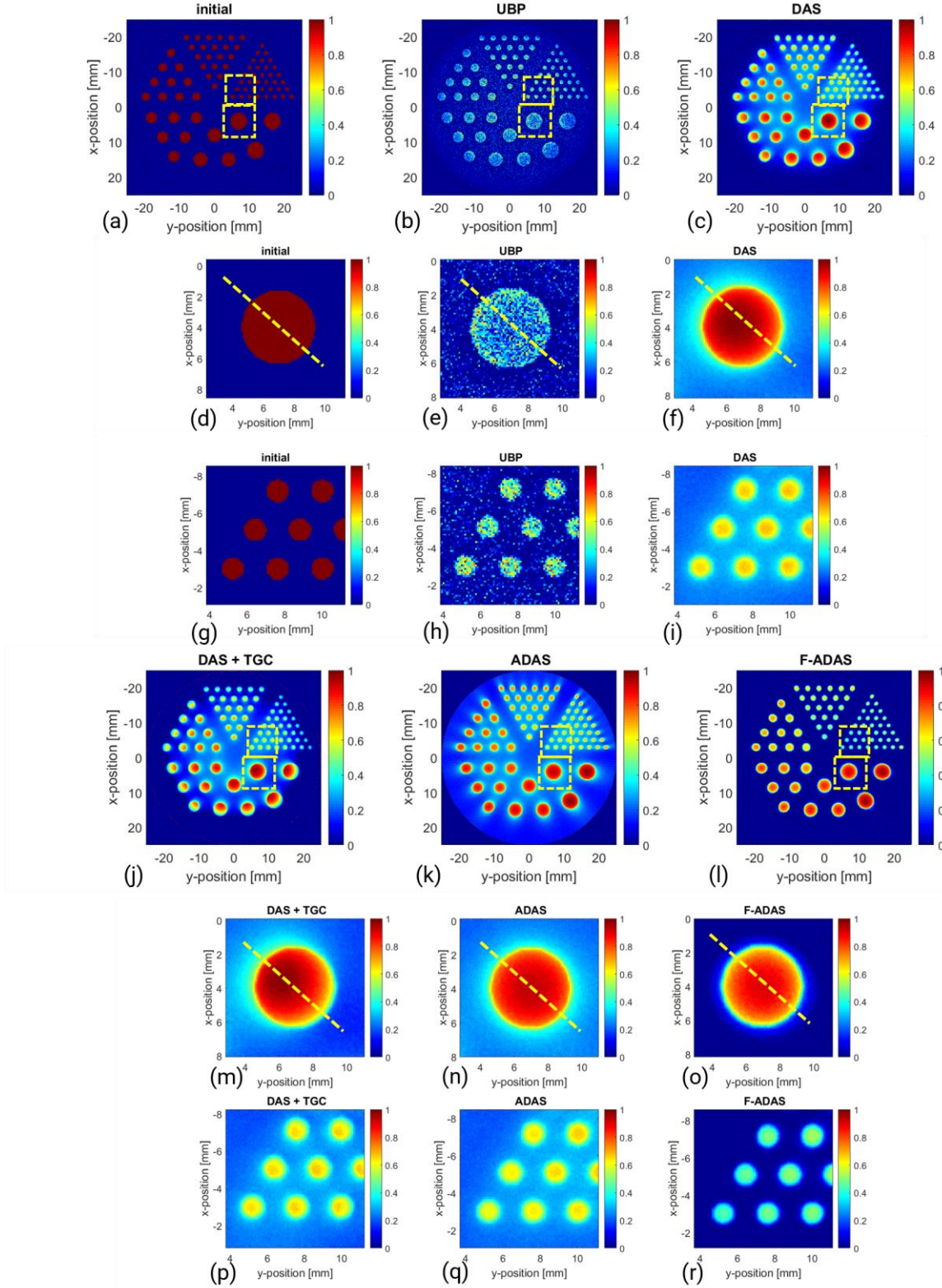


Fig. 4: Numerical results of the reconstructed image. (a) Initial Derenzo phantom. (b, c) are reconstructed images with UBP and DAS, respectively. zoomed areas of dashed lines in (a, b, c) are shown in (d, e, f, g, h, i), respectively. (j, k, l) are reconstructed images with DAS+TGC, ADAS, and F-ADAS, respectively, and indicated dashed lines are shown in (m, n, o, p, q, r), respectively.

reseau of this, is consistence of the ramp filter in UBP algorithm. Reconstructed image with ADAS and F-ADAS are more uniform than others (see in Fig. 6(c)). Moreover, due to the band limited transducer, DC values are removed from all

images and the results of ADAS and F-ADAS are same. The intensity of left edge for all reconstruction algorithms are the same, but for right edge for ADAS/F-ADAS algorithms it is the same as right edge. Quantitative values of all

reconstruction algorithms are indicated in Table 2. In calculation of EUI in this case, we consider both of peaks at the edges of the considered phantom in the indicated 1D-Line (see zoomed profiles in Fig. 6(b, c)).

It can be concluded from Table 2, that uniformity index of the reconstructed image obtained with ADAS and F-ADAS is 1, which means that uniformity with these algorithms is at its maximum value.

TABLE 2: Quantitative values (EUI) for 2.25 MHz, 80% bandwidth and 25 dB SNR level scenario.

Algorithm	UBP	DAS	DAS+TGC	ADAS	F-ADAS
EUI	0.998	0.923	0.908	1	1

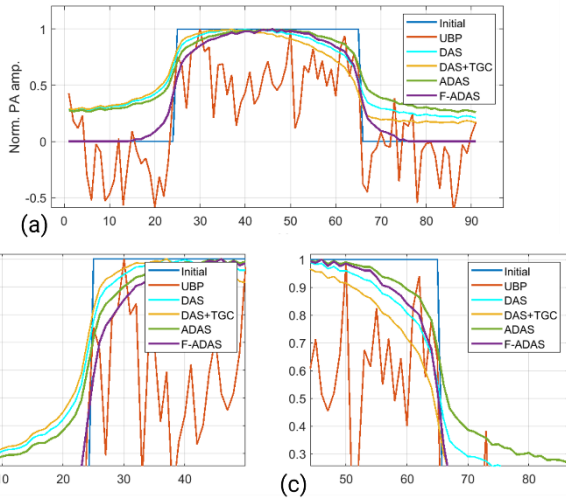


Fig. 5: 1D-line of PA amplitude of yellow dashed line in Fig. 4(d, e, f, j, k, l) for infinite bandwidth and 25 dB SNR level scenario. (b, c) are the zoomed left and right edges of the reconstructed 1D profile, respectively.

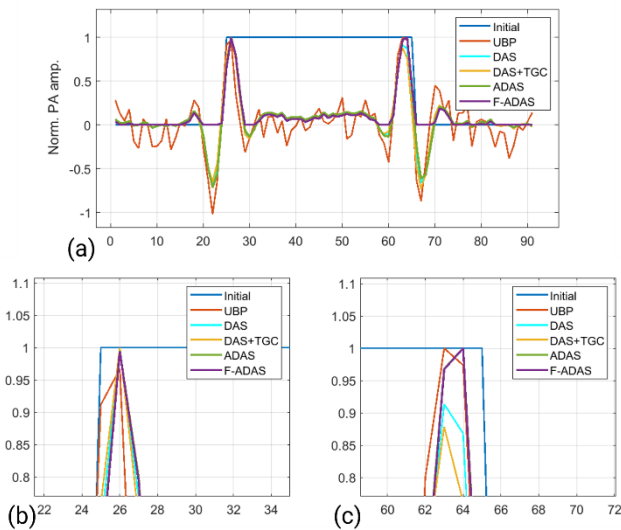


Fig. 6: (a) 1D-line of PA amplitude of yellow dashed line in Fig. 4(d, e, f, m, n, o) for 2.25 MHz, 80% bandwidth and 25 dB SNR level scenario. (b, c) are the zoomed left and right edges of the

reconstructed 1D profile, respectively. The intensity of left edge for all reconstruction algorithms are the same, but for right edge for ADAS/F-ADAS algorithms it is the same as right edge.

IV. DISCUSSION AND CONCLUSION

In this paper, we proposed ADAS and F-ADAS reconstruction algorithm for circular scanning photoacoustic tomography geometries with point-like detectors. The results indicated that using ADAS can help to improve edge uniformity index (EUI) to 1, which is its maximum value in 2.25 MHz, 80% bandwidth scenario and improves EUI by 17.3-39% in infinite bandwidth scenario. Moreover, in infinite bandwidth scenario by adding a DC filter to the processed signal, DC term in reconstructed image can be removed, which is proposed in F-ADAS. Based on the results a uniform reconstructed image can be achieved by using the proposed method (ADAS/F-ADAS). ADAS/F-ADAS have the advantages of low computational cost of the mentioned conventional algorithms (DAS and UBP) while yielding higher quality images which means ADAS can be an alternative to conventional methods.

REFERENCES

- [1] M. Xu and L. V. Wang, "Photoacoustic imaging in biomedicine," *Review of scientific instruments*, vol. 77, no. 4, p. 041101, 2006.
- [2] A. DiSpirito, 3rd, T. Vu, M. Pramanik, and J. Yao, "Sounding out the hidden data: A concise review of deep learning in photoacoustic imaging," *Exp Biol Med (Maywood)*, vol. 246, no. 12, pp. 1355-1367, Jun 2021, doi: 10.1177/15353702211000310.
- [3] S. Hakakzadeh and Z. Kavehvash, "Blind Angle and Angular Range Detection in Planar and Limited-View Geometries for Photoacoustic Tomography," in *2021 29th Iranian Conference on Electrical Engineering (ICEE)*, 18-20 May 2021 2021, pp. 922-926, doi: 10.1109/ICEE52715.2021.9544114.
- [4] S. Hakakzadeh and Z. Kavehvash, "Image Quality Equations for Focused Transducer in Circular Photoacoustic Computed Tomography," in *2021 29th Iranian Conference on Electrical Engineering (ICEE)*, 18-20 May 2021 2021, pp. 917-921, doi: 10.1109/ICEE52715.2021.9544449.
- [5] Soheil Hakakzadeh, Mohammadreza Amjadian, Yachao Zhang, Seyed Masood Mostafavi, and Z. K. a. L. Wang, "Unipolar Adapted Delay and Sum Algorithm and Power Mean to Variance Factor for Photoacoustic Tomography " 2022.
- [6] S. Na and L. V. Wang, "Photoacoustic computed tomography for functional human brain imaging [Invited]," *Biomed. Opt. Express*, vol. 12, no. 7, 2021, doi: 10.1364/boe.423707.
- [7] M. Heijblom, W. Steenberg, and S. Manohar, "Clinical Photoacoustic Breast Imaging: The Twente experience," *IEEE Pulse*, vol. 6, no. 3, pp. 42-46, 2015, doi: 10.1109/MPUL.2015.2409102.
- [8] A. P. Rao, N. Bokde, and S. Sinha, "Photoacoustic Imaging for Management of Breast Cancer: A Literature Review and Future Perspectives," *Applied Sciences*, vol. 10, no. 3, p. 767, 2020, doi: 10.3390/app10030767.
- [9] H.-P. Brecht, R. Su, M. Fronheiser, S. Ermilov, A. Conjusteau, and A. Oraevsky, "Whole-body three-dimensional optoacoustic tomography system for small animals," *Journal of Biomedical Optics*, vol. 14, no. 6, p. 064007, 2009.
- [10] S. Hakakzadeh *et al.*, "Multi-Angle Data Acquisition to Compensate Transducer Finite Size in Photoacoustic Tomography," 2021 2021.
- [11] A. Hauptmann *et al.*, "Model-Based Learning for Accelerated, Limited-View 3-D Photoacoustic Tomography," *IEEE Trans Med*

- Imaging*, vol. 37, no. 6, pp. 1382-1393, Jun 2018, doi: 10.1109/TMI.2018.2820382.
- [12] S. Hakakzadeh *et al.*, "Finite Transducer Size Compensation in Two-Dimensional Photoacoustic Computed Tomography," in *2021 IEEE International Ultrasonics Symposium (IUS)*, 11-16 Sept. 2021 2021, pp. 1-3, doi: 10.1109/IUS52206.2021.9593757.
- [13] A. Dima and V. Ntziachristos, "In-vivo handheld optoacoustic tomography of the human thyroid," *Photoacoustics*, vol. 4, no. 2, pp. 65-69, 2016/06/01/ 2016,
- [14] M. Xu and L. V. Wang, "Universal back-projection algorithm for photoacoustic computed tomography," *Physical Review E*, vol. 71, no. 1, 2005, doi: 10.1103/physreve.71.016706.
- [15] G. Matrone, A. S. Savoia, G. Caliano, and G. Magenes, "The Delay Multiply and Sum Beamforming Algorithm in Ultrasound B-Mode Medical Imaging," *IEEE Transactions on Medical Imaging*, vol. 34, no. 4, pp. 940-949, 2015, doi: 10.1109/tmi.2014.2371235.
- [16] Soheil Hakakzadeh, Seyed Masood Mostafavi, Mohammadreza Amjadian, and Zahra Kavehvash, "Adapted Coherent Weighting in Photoacoustic Tomography," presented at the 28th National and 6th International Iranian Conference on Biomedical Engineering (ICBME).
- [17] M. A. Araque Caballero, J. Gateau, X.-L. Dean-Ben, and V. Ntziachristos, "Model-Based Optoacoustic Image Reconstruction of Large Three-Dimensional Tomographic Datasets Acquired With an Array of Directional Detectors," *IEEE Transactions on Medical Imaging*, vol. 33, no. 2, pp. 433-443, 2014, doi: 10.1109/tmi.2013.2286546.